

## SUMMARY TABLE: CORE MATHEMATICS GRADE 10

### Sub-Strand 2.5: Area of Polygons — Teacher Reference (Pre-filled)

| SUMMARY TABLE: CORE MATHEMATICS GRADE 10 |                                                                                                                                                                                                                                 |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sub-Strand                               | Sub-Strand 2.5: Area of Polygons                                                                                                                                                                                                |
| Driving Question                         | DRIVING QUESTION: How can we calculate the area of any polygon — no matter how many sides or how irregular — so that we can fairly divide the Kericho community land plot and solve other real Kenyan land and design problems? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                                              | What did I observe?                                                                                                                                                                                                                                                                                                                                                                         | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                            | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lesson 1</b><br><b>Anchoring the Phenomenon — The Kericho Land Plot: How Much Land Is There, and Can We Share It Fairly?</b> | A scaled drawing of the Kericho seven-sided land plot — an irregular heptagon with seven unequal sides and no right angles. Learners observed that the plot cannot be measured using any single standard formula (rectangle, square, circle) and that the community's inability to calculate the area threatens fair land distribution and food security for three Kenyan farming families. | The area of a triangle is calculated using $A = \frac{1}{2}bh$ , where $h$ is the PERPENDICULAR height from the base to the opposite vertex — NOT the slant side. A heptagon can be decomposed into five triangles by drawing diagonals from a single vertex. The total area of the polygon equals the SUM of the areas of all five triangles. This makes the triangle the fundamental 'building block' for finding the area of any polygon. | If we draw diagonals from one vertex of the Kericho heptagon to all non-adjacent vertices, we divide the plot into five triangles. The total area of the land plot equals the sum of the areas of these five triangles, each calculated using $A = \frac{1}{2}bh$ . This approach — decomposing any polygon into triangles — is a general strategy that could work for a polygon with any number of sides, which means we have the beginning of a method to help the three farming families near Kericho divide their land fairly. |
| <b>Lesson 2</b><br><b>Quadrilaterals — Parallelograms, Rhombuses, Trapeziums and Kites</b>                                      | Learners measured the base, perpendicular height, and/or diagonals of four quadrilateral cut-outs (parallelogram shaped like an iron-sheet rooftop panel, rhombus, trapezium shaped like a Kericho tea-farm terrace cross-section, and kite). They physically rearranged a parallelogram into a rectangle and dissected a trapezium into two triangles.                                     | Learners derived and applied the four area formulae: parallelogram $A = bh$ ; rhombus $A = \frac{1}{2}d_1d_2$ ; trapezium $A = \frac{1}{2}(a+b)h$ ; kite $A = \frac{1}{2}d_1d_2$ . They learned that the perpendicular height (not the slant side) must be used in the parallelogram and trapezium formulae, and that the rhombus and kite share a formula because both are fully determined by their two diagonals.                         | Learners explained that each quadrilateral formula is derived by linking the shape to a rectangle or pair of triangles whose areas are already known. They applied the formulae to calculate the floor area of a parallelogram-shaped cattle pen in Naivasha and the cross-sectional area of a trapezoidal tea terrace in Kericho, and they updated their decomposition model of the seven-sided Kericho community land plot to include quadrilateral sub-regions alongside triangles.                                             |

|                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lesson 3</b><br><b>Trigonometric Area of Triangles — When the Height Is Hidden</b>    | <p>Learners observed that a triangle with two known sides (5 m and 7 m) and a known included angle (<math>58^\circ</math>) has a 'hidden' perpendicular height that cannot be measured directly. By drawing the height and applying the sine ratio, they saw that <math>h = a \sin C</math>, which when substituted into <math>A = \frac{1}{2}bh</math> produces the formula <math>A = \frac{1}{2}ab \sin C</math>. They verified this formula numerically using both a scientific calculator and mathematical tables, confirming that the area of the Kisumu solar panel triangle is approximately <math>14.84 \text{ m}^2</math>.</p> | <p>The formula <math>A = \frac{1}{2}ab \sin C</math> gives the area of any triangle when two side lengths (<math>a</math> and <math>b</math>) and the included angle (<math>C</math>) are known — even when the perpendicular height cannot be measured. This formula is derived directly from <math>A = \frac{1}{2}bh</math> by substituting <math>h = a \sin C</math>, so it is not a new rule but an extension of the foundational triangle area formula. A sine value can be found using either a scientific calculator or mathematical tables, and both methods should give the same result.</p>        | <p>The Kericho land plot is decomposed into five triangles. Some of these triangles will have a perpendicular height that is difficult or impossible to measure directly from the boundary drawing — but the side lengths and angles between them CAN be read from the scaled diagram. The formula <math>A = \frac{1}{2}ab \sin C</math> means that EVERY triangle in the decomposed polygon can now be solved, regardless of whether its height is accessible. This removes the last remaining barrier to calculating the total area of the Kericho plot and dividing it fairly among the three farming families.</p> |
| <b>Lesson 4</b><br><b>Heptagons — Decomposing the Kericho Land Plot into Triangles</b>   | <p>Learners used rulers and triangular card cut-outs to decompose a scaled drawing of the seven-sided Kericho community land plot into five non-overlapping triangles, measuring each triangle's base and perpendicular height.</p>                                                                                                                                                                                                                                                                                                                                                                                                     | <p>The area of a heptagon equals the sum of the areas of five non-overlapping triangles formed by drawing diagonals from one interior vertex: <math>A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \frac{1}{2}b_3h_3 + \frac{1}{2}b_4h_4 + \frac{1}{2}b_5h_5</math>. In general, any <math>n</math>-gon can be decomposed into <math>(n - 2)</math> triangles.</p>                                                                                                                                                                                                                                              | <p>Any polygon — no matter how many sides — can be broken into <math>(n - 2)</math> non-overlapping triangles, so its area can always be calculated using the triangle area formula. This means the Kericho community can calculate the exact area of their irregular seven-sided plot and divide it fairly among the three farming families.</p>                                                                                                                                                                                                                                                                      |
| <b>Lesson 5</b><br><b>Octagons — Decomposing the Mombasa Courtyard</b>                   | <p>Students observed that a regular octagon drawn to scale (side = 3 m) can be divided into eight congruent isosceles triangles by drawing lines from the centre <math>O</math> to each vertex. They measured the central angle of each triangle as <math>45^\circ</math> (confirming <math>360^\circ \div 8 = 45^\circ</math>) and measured the apothem from the scale diagram to use in the triangle area calculation.</p>                                                                                                                                                                                                            | <p>Students learned that the area of a regular octagon can be determined by two equivalent methods: (a) decomposing it into 8 isosceles triangles and summing their areas using <math>A_{\text{triangle}} = \frac{1}{2} \times \text{base} \times \text{height}</math>; and (b) applying the direct formula <math>A = 2(1 + \sqrt{2})s^2</math>. They learned that the formula is derived from the decomposition method, not a separate rule. They also learned how to calculate tile quantities (including 10% wastage allowance) and total material cost in Kenyan Shillings from an area calculation.</p> | <p>Students explained that the regular octagon's area formula <math>A = 2(1 + \sqrt{2})s^2</math> emerges directly from multiplying the area of one isosceles triangle by eight, and that this decomposition-into-triangles strategy is the same strategy used for every polygon in the lesson sequence (triangles, quadrilaterals, heptagons, octagons). They explained that the same principle — decomposing any polygon into triangles from an interior point — will work for the irregular seven-sided Kericho community land plot, regardless of its irregular side lengths and angles.</p>                       |
| <b>Lesson 6</b><br><b>Irregular Polygons — Triangulation of the Nakuru School Shamba</b> | <p>Learners worked with a scaled diagram of a five-sided irregular polygon representing the boundary of a school shamba near Nakuru. They drew three non-overlapping triangles inside the pentagon by connecting vertex <math>A</math> to all non-adjacent vertices, measured each triangle's base and perpendicular height using rulers and set squares, computed each triangle's area using <math>A = \frac{1}{2}bh</math>,</p>                                                                                                                                                                                                       | <p>Any irregular polygon with <math>n</math> sides can be divided into exactly <math>(n-2)</math> non-overlapping triangles by drawing diagonals from one reference vertex to all non-adjacent vertices. The total area of the polygon equals the sum of the areas of these triangles (<math>A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \dots + \frac{1}{2}b_{\{n-2\}}h_{\{n-2\}}</math>). When working from a scale diagram, areas must be converted using the</p>                                                                                                                                         | <p>The triangulation method works for ANY polygon — regular or irregular, regardless of the number of sides — because every polygon can be decomposed into non-overlapping triangles, and we already know how to find the area of a triangle. This means the method discovered today can be applied directly to the seven-sided Kericho community plot: by drawing five diagonals</p>                                                                                                                                                                                                                                  |

|                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                             | converted from $\text{cm}^2$ to $\text{m}^2$ using the squared scale factor ( $1 \text{ cm}^2 = 16 \text{ m}^2$ ), and summed the three triangle areas to find the total shamba area. Groups posted their totals on the Evidence Wall and observed that different groups obtained different answers for the same shape.                                                                                | square of the linear scale factor (if 1 cm represents $k$ metres, then $1 \text{ cm}^2$ represents $k^2$ square metres). Discrepancies between group results arise from measurement error or incorrect perpendicular height identification, not from the method itself. The coordinate (shoelace) formula provides an alternative verification method used by professional surveyors.                 | from one vertex, we create five triangles whose areas sum to the total plot area, giving the village elder the precise measurement needed to divide the land fairly among the three farming families.                                                                                                                                                                                                                                                  |
| <b>Lesson 7</b><br><b>Composite Polygons and Real-Life Application: Solving the Kericho Land-Division Problem</b>           | Learners solved two composite polygon problems (an L-shaped school building footprint and a pentagonal community garden with a rectangular section removed) using both additive and subtractive area strategies, recording their working in a structured three-column evidence table and confirming that two different decompositions of the L-shape yield the same total area.                        | Learners consolidated the principle that the area of any composite polygon equals the sum of its component standard-polygon areas, or a whole area minus a removed section — expressed as $\text{Area}(\text{composite}) = \sum \text{Area}(\text{components}) \pm \text{Area}(\text{removed section})$ — and that this principle applies regardless of which valid decomposition strategy is chosen. | Learners applied their full polygon-area toolkit (Lessons 1–7) to the anchoring phenomenon: they decomposed the Kericho seven-sided community plot into standard shapes, calculated the total area, divided it equally into three parts for the Wanjiku, Otieno, and Kosgei farming families, and produced a labelled diagram with a written justification of the fair division, preparing a complete solution for community presentation in Lesson 8. |
| <b>Lesson 8</b><br><b>Model Building, DQB Resolution &amp; Final Assessment — Solving the Kericho Land Division Problem</b> | Groups presented their annotated decomposition diagrams of the seven-sided Kericho heptagonal plot, showing how they split it into triangles and quadrilaterals, computed sub-areas using known formulae, and proposed boundary lines dividing the total area equally among three farming families. Multiple valid decomposition strategies were compared and shown to produce equivalent total areas. | Any polygon — no matter how many sides or how irregular — can be decomposed into triangles and quadrilaterals whose areas can be calculated using known formulae. The total area equals the sum of the parts. This principle, together with the formulae for all polygon types, gives us a complete toolkit to solve any land-area problem in Kenya or beyond.                                        | We resolved the driving question: the Kericho community land plot has a calculable total area found by decomposition, and it can be divided fairly among three families by allocating boundary lines so that each section equals exactly one-third of the total area. The 'Area of Polygons Toolkit' poster and DQB resolution show that every question raised in Lesson 1 has now been answered through eight lessons of investigation.               |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

**Licensing & Copyright:** © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.